Asignment4- Univariate Analysis

Andre King

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```
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.1.4     v readr     2.1.5
## v forcats   1.0.0     v stringr   1.5.1
## v ggplot2   3.5.2     v tibble    3.2.1
## v lubridate 1.9.4     v tidyr     1.3.1
## v purrr     1.0.2
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(haven)
library(dplyr)
library(ggplot2)


setwd("/Users/senorking/Desktop/Fall2025/Data333") # modify as needed
```

Import Data

```
students <- read.csv("Students and Social Media.csv")
str(students)
```

```
## 'data.frame':    705 obs. of  13 variables:
##  $ Student_ID                  : int  1 2 3 4 5 6 7 8 9 10 ...
##  $ Age                         : int  19 22 20 18 21 19 23 20 18 21 ...
##  $ Gender                      : chr  "Female" "Male" "Female" "Male" ...
##  $ Academic_Level              : chr  "Undergraduate" "Graduate" "Undergraduate" "High School" ...
##  $ Country                     : chr  "Bangladesh" "India" "USA" "UK" ...
##  $ Avg_Daily_Usage_Hours       : num  5.2 2.1 6 3 4.5 7.2 1.5 5.8 4 3.3 ...
##  $ Most_Used_Platform          : chr  "Instagram" "Twitter" "TikTok" "YouTube" ...
##  $ Affects_Academic_Performance: chr  "Yes" "No" "Yes" "No" ...
##  $ Sleep_Hours_Per_Night       : num  6.5 7.5 5 7 6 4.5 8 6 6.5 7 ...
##  $ Mental_Health_Score         : int  6 8 5 7 6 4 9 6 7 7 ...
##  $ Relationship_Status         : chr  "In Relationship" "Single" "Complicated" "Single" ...
##  $ Conflicts_Over_Social_Media : int  3 0 4 1 2 5 0 2 1 1 ...
##  $ Addicted_Score              : int  8 3 9 4 7 9 2 8 5 4 ...
```

```r
head(students)
```

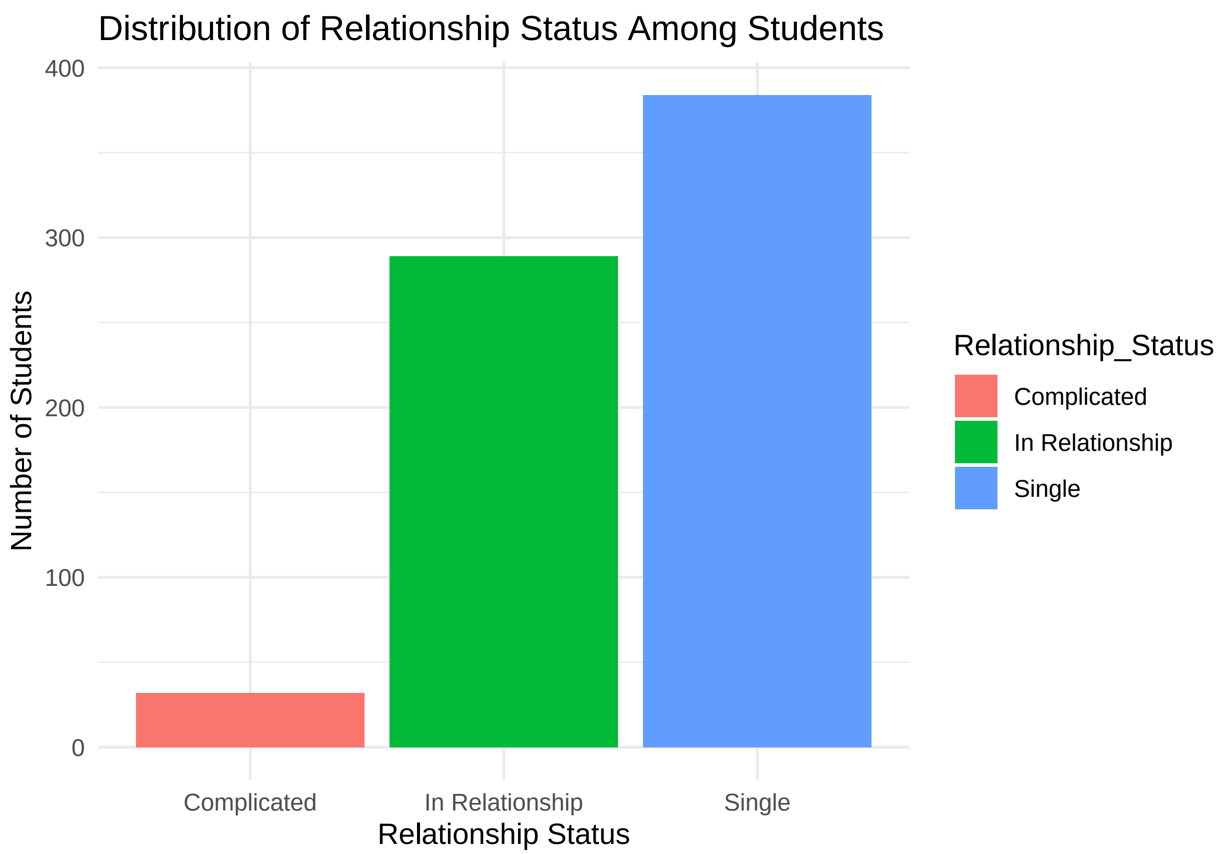
```
##   Student_ID Age Gender Academic_Level    Country Avg_Daily_Usage_Hours
## 1          1  19 Female  Undergraduate Bangladesh                   5.2
## 2          2  22   Male       Graduate      India                   2.1
## 3          3  20 Female  Undergraduate        USA                   6.0
## 4          4  18   Male    High School         UK                   3.0
## 5          5  21   Male       Graduate     Canada                   4.5
## 6          6  19 Female  Undergraduate  Australia                   7.2
##   Most_Used_Platform Affects_Academic_Performance Sleep_Hours_Per_Night
## 1          Instagram                          Yes                   6.5
## 2            Twitter                           No                   7.5
## 3             TikTok                          Yes                   5.0
## 4            YouTube                           No                   7.0
## 5           Facebook                          Yes                   6.0
## 6          Instagram                          Yes                   4.5
##   Mental_Health_Score Relationship_Status Conflicts_Over_Social_Media
## 1                   6     In Relationship                           3
## 2                   8              Single                           0
## 3                   5         Complicated                           4
## 4                   7              Single                           1
## 5                   6     In Relationship                           2
## 6                   4         Complicated                           5
##   Addicted_Score
## 1              8
## 2              3
## 3              9
## 4              4
## 5              7
## 6              9
```

There are 705 cases and 13 variables in the data imported for ananlysis.

Task 1

The variable Relationship_Status is a categorical variable that describes each student's romantic relationship status. It measures which relationship category the student identifies with. This variable includes the following categories: Single, In Relationship, Complicated.

```r
ggplot(students, aes(x = Relationship_Status, fill = Relationship_Status)) +
  geom_bar() +
  labs(
    title = "Distribution of Relationship Status Among Students",
    x = "Relationship Status",
    y = "Number of Students"
  ) +
  theme_minimal()
```

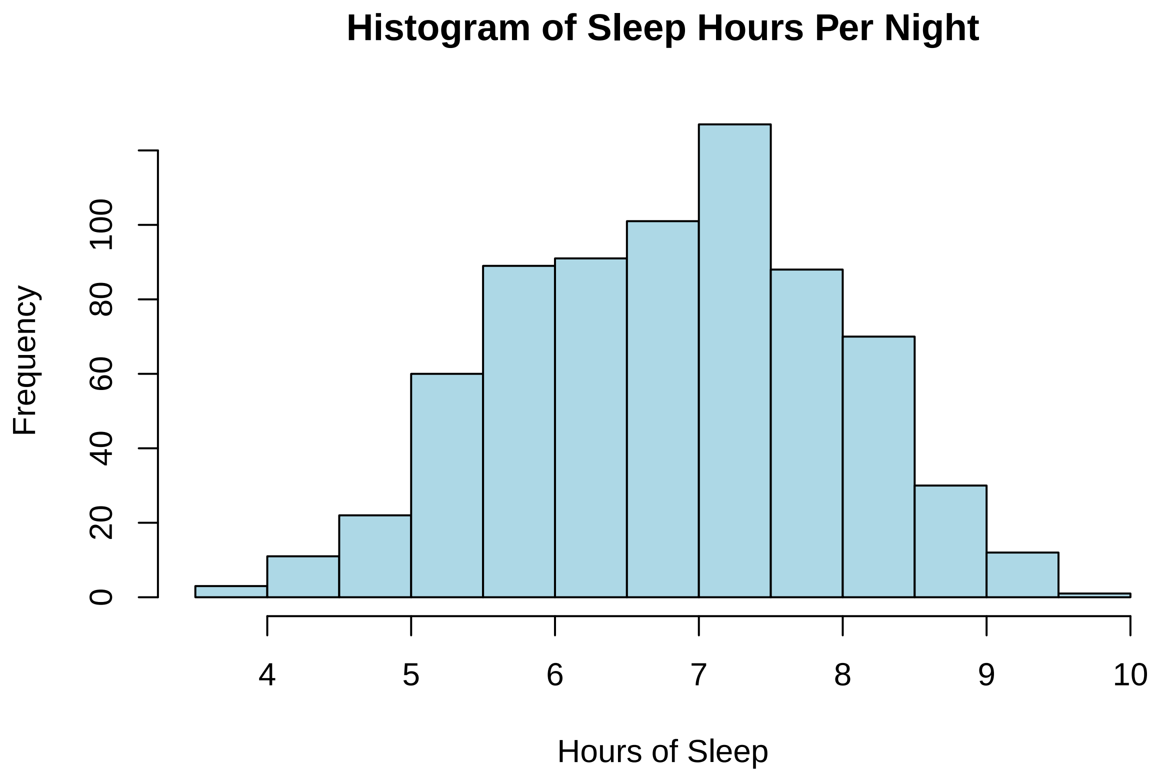


The bar chart shows the distribution of relationship status among students in the dataset. The 'Single' category is the most common, with approximately 380 students, making it the dominant group in the dataset. The 'In Relationship' category is the second largest, with around 290 students, indicating that a substantial portion of students are currently in a relationship. The 'Complicated' category is much smaller, with only about 30 students. This makes it the least popular group and a relatively rare category compared with the other two relationship statuses.

Task 2

The variable Sleep_Hours_Per_Night is a numeric variable that measures how many hours of sleep each student gets on a typical night. The metric is hours per night and is recorded as a continuous number, for example: 4.5 hours, 7 hours, 8 hours. This variable helps to show students' sleep patterns and how much sleep they are getting on average.

```r
hist(students$Sleep_Hours_Per_Night,
     main = "Histogram of Sleep Hours Per Night",
     xlab = "Hours of Sleep",
     col = "lightblue")
```

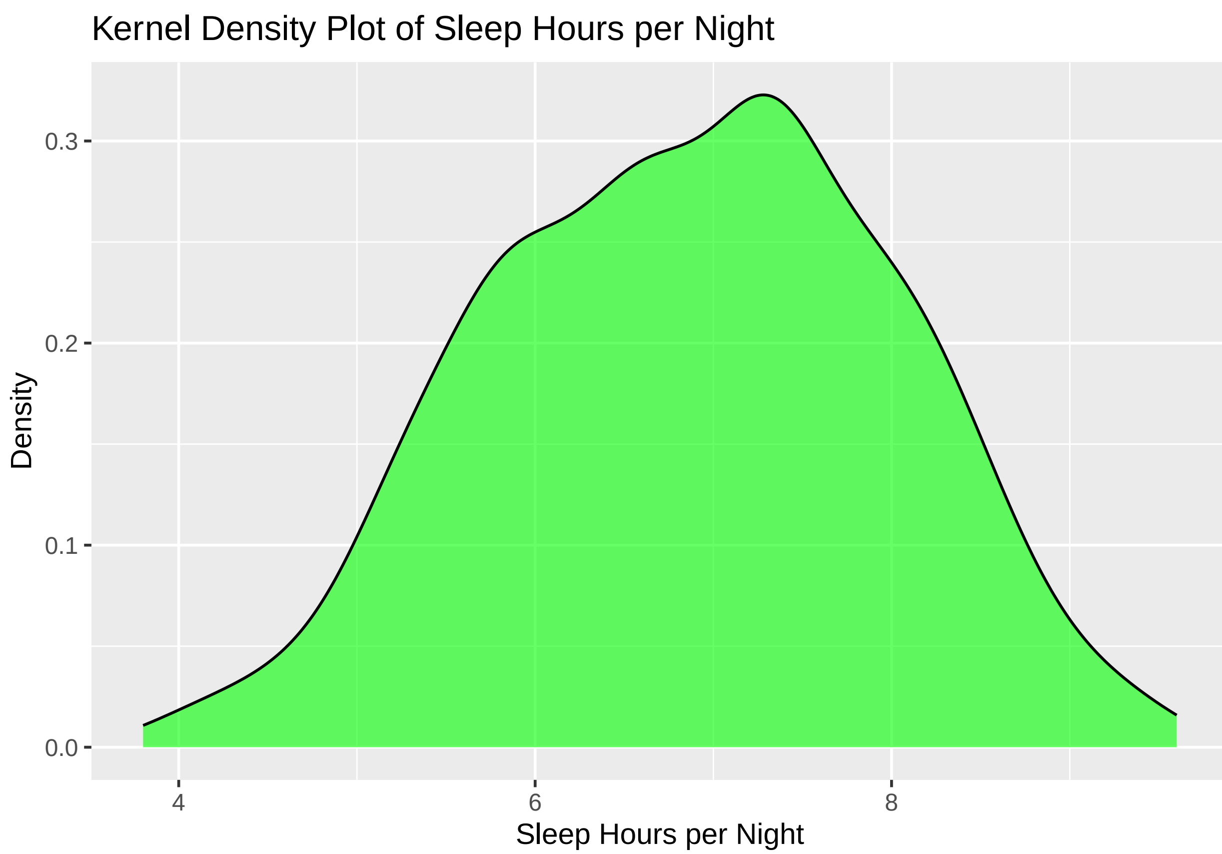


The histogram shows a roughly bell-shaped distribution, this means that the data is approximately normal. Most of the students sleep for about 5.5 to 8 hours with fewer observations near the extremes. The minimum is about 3 hours and the maximum is about 10 hours. The peak of the data is about 7.5 hours

Task 3

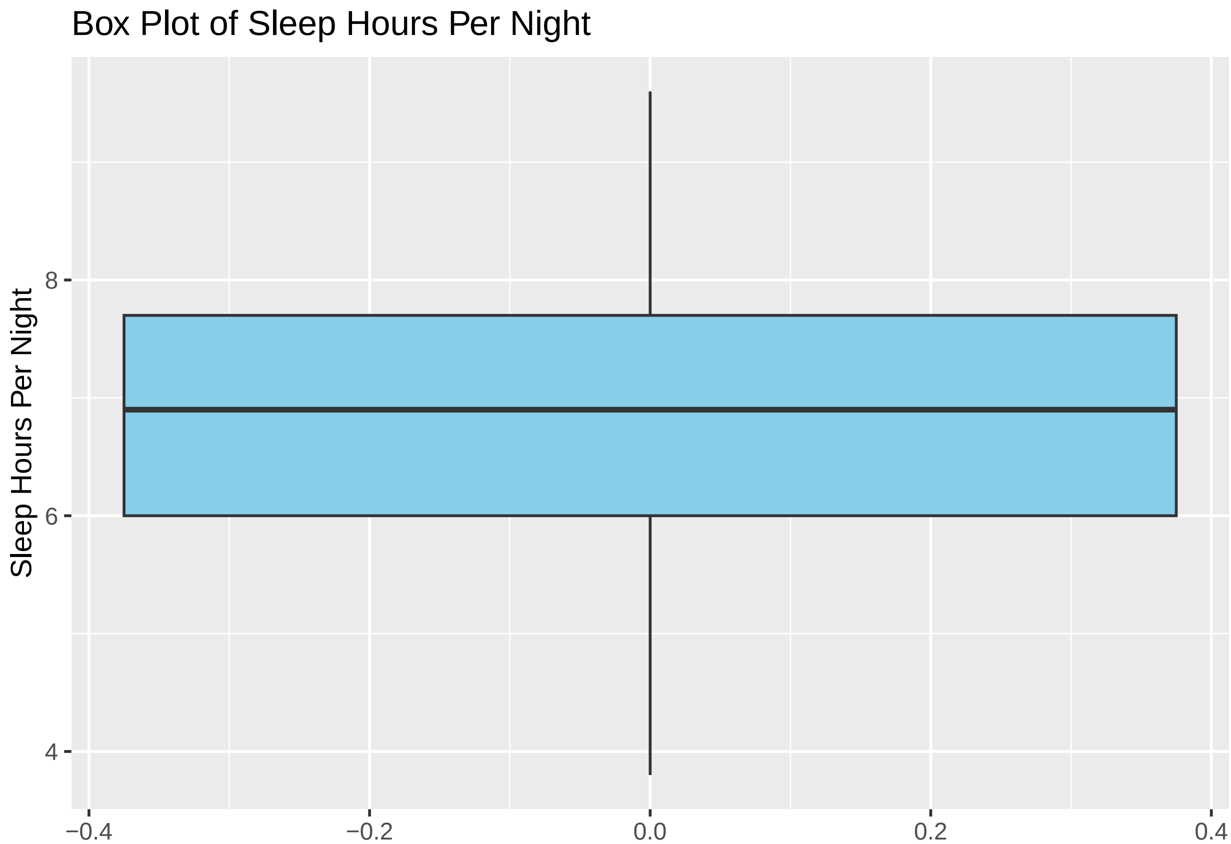
Kernel density plot of Sleep_Hours_Per_Night. A kernel density plot removes the blocky appearance and gives a continuous curve, making the overall shape easier to interpret. The density curve helps you see whether the data is close to normal, skewed, or multi-peaked without the distraction.

```
ggplot(students, aes(x = Sleep_Hours_Per_Night)) +
  geom_density(fill = "green", alpha = 0.6) +
  labs(
    title = "Kernel Density Plot of Sleep Hours per Night",
    x = "Sleep Hours per Night",
    y = "Density" )
```



Task 4

```
ggplot(students, aes(y = Sleep_Hours_Per_Night)) +
  geom_boxplot(fill = "skyblue") +
  labs(
    title = "Box Plot of Sleep Hours Per Night",
    y = "Sleep Hours Per Night")
```



Task 5

The kernel density plot is the best for visualizing this numeric variable because it gives the smoothest and clearest picture of the overall distribution. It shows the peak, how the data is spread out, and whether the data is symmetric without being affected by rectangular width like a histogram. The histogram is also great to use because it show the number of student for each time slots and using more rectangles will give better view.